

BMW Group Delivers First-Class Generative AI–Based Cloud Optimization Assistant Using AWS

Contact Us

“Using Amazon Bedrock, we’ve been able to scale our cloud governance, reduce costs and time to market, and provide a better service for our customers. All of this is helping us deliver the secure, first-class digital experiences that people across the world expect from BMW.”

Dr. Jens Kohl
Head of Offboard Architecture, I

 Hi, I can connect you with an AWS representative or answer questions you have on AWS. 

One of the best-known names in the auto experiences to millions of customers since groundbreaking In-Console Cloud Assistant (ICCA) solution in collaboration with Amazon Web Services (AWS). Part of the company’s infrastructure optimization program, the ICCA is empowering BMW Group I generative artificial intelligence (AI) to boost performance and efficiency. With access to large

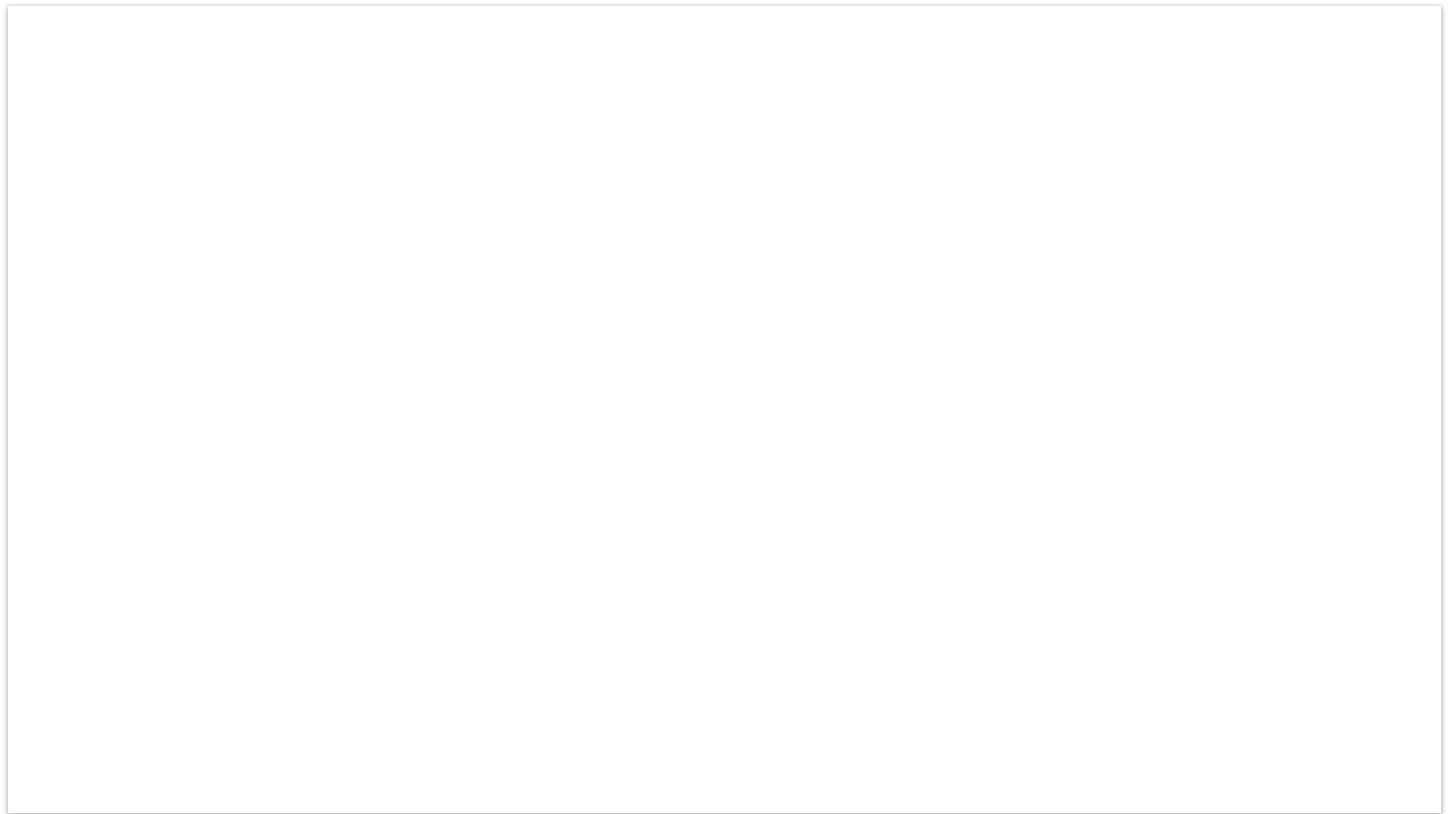
 Need more info? Highlight any text to get an explanation generated with AWS generative AI. 





2

(LLMs) in Amazon Bedrock—and with data encrypted in transit and at rest—BMW Group can securely deliver high-quality connected mobility solutions to motorists around the world



A Premium Experience That Moves with the Times

BMW Group is a name that needs little introduction. A manufacturer of luxury vehicles, its brands—which include BMW, MINI, and Rolls-Royce—are recognized across the globe. In its more than 100-year history it hasn’t stood still. And today, as automotive manufacturers become technology-driven innovators, it continues to lead the way.

The business has looked to AWS to help it make this transition. From migrating its Cloud Data Hub to AWS for greater scale and agility. To using serverless analytics to personalize the digital in-car experience for customers. The AWS Cloud helps its teams realize BMW Group’s mission: “to become the world’s leading provider of premium products and services for individual mobility.”

Scale + Intelligence + Optimization = Innovation

AI has been a critical component of BMW Group’s customer-centric approach for many years. From predictive maintenance of vehicle components and delivery of real-time traffic information to developing Highly automated and autonomous driving functionality—AI has transformed driving for BMW customers.

Jens Kohl, head of offboard architecture at BMW Group, and his team need to ensure the experiences they create can be scaled across 22.3 million vehicles. “Our connected vehicle backend hosted on AWS processes 14.3 billion requests and 145 terabytes of traffic a day, as in March of 2024,” he says. “AWS helps us scale almost limitlessly in real time. It’s good to know we can meet demand as the number of customers engaging with our services grows.”





But with over 450 DevOps teams working across more than 450 AWS accounts to produce business-critical applications, optimizing cloud infrastructure was a challenge. BMW Group worked with the [AWS Generative AI Innovation Center](#)—which connects organizations with AWS science and strategy experts with deep experience in AI, machine learning, and generative AI techniques—to find a solution.

ICCA—the conversational generative AI-based assistant they created—uses Amazon Bedrock, a new service from AWS that makes it easy to build and scale generative AI applications. With the ability to understand natural language requests, it helps BMW Group DevOps teams accelerate the optimization process by monitoring performance, identifying bottlenecks, and detecting opportunities for improvement across accounts. The access to a range of LLMs on Amazon Bedrock is key. “We can switch between different large language models, using the ones that work best for specific use cases,” says Kohl. “It’s great to have flexibility and freedom of choice.”



Trust and Competitive Advantage

The beauty of Amazon Bedrock, continues Kohl, is that his team can use it like any other AWS service or component. This means no data moves outside of the BMW Cloud Room, which is hosted on AWS. “Our highest value asset is our customers’ trust,” he says. “Using AWS, we know our customer data is exactly where it should be and protected at all times.”

Harnessing the potential of generative AI on AWS while maintaining customer trust is proving a competitive advantage for BMW Group. “Using Amazon Bedrock, we’ve been able to scale our cloud governance, reduce costs and time to market, and provide a better service for our customers,” says Kohl. “All of this is helping us deliver the secure, first-class digital experiences that people across the world expect from BMW.”

[Read how BMW Group DevOps streamlined cloud optimization with generative AI on AWS](#)

Leading Cloud Innovators in Europe

Learn how leading organizations in Europe across industries trust AWS to drive innovation at every level of their business.

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.

Contact Sales

Learn About AWS

- What Is AWS?
- What Is Cloud Computing?
- AWS Accessibility
- AWS Inclusion, Diversity & Equity
- What Is DevOps?
- What Is a Container?
- What Is a Data Lake?
- What is Artificial Intelligence (AI)?
- What is Generative AI?
- What is Machine Learning (ML)?
- AWS Cloud Security
- What's New
- Blogs
- Press Releases

Resources for AWS

- Getting Started
- Training and Certification
- AWS Solutions Library
- Architecture Center
- Product and Technical FAQs
- Analyst Reports
- AWS Partners

Developers on AWS

- Developer Center
- SDKs & Tools
- .NET on AWS
- Python on AWS
- Java on AWS
- PHP on AWS
- JavaScript on AWS

Help

- Contact Us
- Get Expert Help
- File a Support Ticket
- AWS re:Post
- Knowledge Center
- AWS Support Overview
- Legal
- AWS Careers



Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language

- عربي |
- Bahasa Indonesia |
- Deutsch |
- English |



Español |
Français |
Italiano |
Português |
Tiếng Việt |
Türkçe |
Русский |
ไทย |
日本語 |
한국어 |
中文 (简体) |
中文 (繁體)

Privacy
|
Accessibility
|
Site Terms
|
Cookie Preferences
|

© 2024, Amazon Web Services, Inc. or its affiliates. All rights reserved.

